

## 第二章 函数的导数 (Derivative)

## 2.1 导数的定义 (The Definition of Derivative)



## Multiple Choice Questions

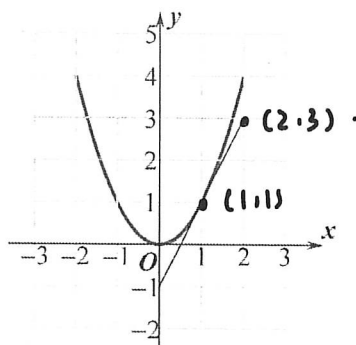
1. Which of the following values is the average rate of change of  $f(x) = \sqrt{x+2}$  over the interval  $[-1, 2]$ ?

- (A) -4 (B)  $-\frac{1}{3}$  (C)  $\frac{1}{3}$  (D) 3

2. Let  $f(x) = 4 - 3x$ . Which of the following is equal to  $f'(0)$ ?

- (A) 4 (B) -4 (C) 3 (D) -3

3. The graph of  $f$  and its tangent line at  $x=1$  is shown below, find  $f'(1) =$  ( ).



- (A) 0 (B)  $\frac{1}{2}$  (C) 1 (D) 2

4. Let  $f(x) = kx^3 - 2x$ , ( $k \neq 0$ ). The average rate of  $f(x)$  over the interval  $[-2, -1]$  is 3, find  $k$ .

- (A)  $\frac{6}{7}$  (B)  $\frac{1}{7}$  (C)  $-\frac{1}{7}$  (D)  $-\frac{5}{7}$

5. **Calculator** Let  $y = \tan^{-1} x - x^2 + 1$ . Find  $\frac{dy}{dx} \big|_{x=0.5} =$

- (A) -12.056 (B) 0 (C) 271.066 (D) 273.066

6. **Calculator** The graph of  $y = \sqrt{x^2 + 1} - 3x$  crosses the  $x$ -axis at one point in the interval  $[0, 1]$ . What is the slope of the graph at this point?

- (A) 0.354 (B) -0.001 (C) -2.666 (D) 0

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{2}{1}$$

$$69k \Rightarrow 3$$

$$\frac{2 - k + 8k - 4}{1}$$

$$7k = 3$$

$$7k = -3$$

$$k = -\frac{3}{7}$$

$$\frac{dy}{dx} = \frac{1}{1+x^2} - 2x$$

$$\tan^{-1} x - x^2 + 1$$

$$-\tan^{-1} x + 2x$$

$$h$$

$$3x = \sqrt{x^2 + 1}$$

$$9x^2 = x^2 + 1$$

$$8x^2 = 1$$

$$x^2 = \frac{1}{8}$$

$$x = \frac{\sqrt{2}}{4}$$

7. The population of insects at time  $t$  days is modeled by the function  $P$  with first derivative  $P'(t) = 0.1t^2 + 10t + 120$ . Which of the following statements is true?

- (A) At time  $t=10$ , the population of insects is 230.  
 (B) At time  $t=10$ , the population of insects is increasing at a rate of 230 insects per day.  
 (C) At time  $t=10$ , the population of insects is decreasing at a rate of 230 insects per day.  
 (D) At time  $t=10$ , the rate of change of the number of insects is increasing at a rate of 230 insects per day.

8. If the line tangent to the graph of a function  $g$  at the point  $(1, 4)$  passes through the point  $(-2, -5)$ , what is the slope of this line?

- (A) 3 (B) -3 (C)  $\frac{1}{3}$  (D)  $-\frac{1}{3}$

### Free Response Questions

9. Let  $f(x) = x^2 - 2x$

(a) Find the average rate of change of  $f$  over  $[1, 3]$ .

(b) Find  $\frac{f(2+h) - f(2)}{h}$ .

(c) Find the instantaneous rate of change of  $f$  at  $x=2$ .

(b):  $\frac{f(2+h) - f(2)}{h} = \frac{(2+h)^2 - 2(2+h) - 4 + 4}{h}$

(c):  $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = 2$

$$= \frac{4h^2 + 4h - 4 + 4}{h} = \frac{4h^2 + 4h}{h} = \frac{h^2 + 2h}{h} = h + 2$$

10. Use the definition of derivative to find the derivative of  $f(x) = x^3$ , then find  $f'(2)$ .

$\frac{f(x+h) - f(x)}{h}$

$= \frac{(x+h)^3 - x^3}{h}$

$= \frac{(x+h)(x+h)(x+h) - x^3}{h}$

$= \frac{(x^2 + 2xh + h^2)(x+h) - x^3}{h}$

$= \frac{x^3 + 2x^2h + x^2h + x^2h + 2xh^2 + h^3 - x^3}{h} = \frac{4x^2h + 2xh^2 + h^3}{h} = 4x^2 + 2xh + h^2$

$f'(x) = \lim_{h \rightarrow 0} (4x^2 + 2xh + h^2)$

$= 4x^2$

$\therefore f'(2) = 16$